nRF52840 BLE Sensor Node

RESET

+3V-BAT
R31 47kΩ
GND
SW3
RST/NRF52840
C6 100nF

+3V-BAT
C8 100nF
C4 100nF

LED_STATE/P1.10/NRF52840
LED1 RED
LED
R39 1kΩ

H1
P0.03 P1.11
AI4 P1.13
AI0 AI5
AI7 AI6
2.54-2*4P立贴针

U5 E73-2G4M08S1C

P1.11	1	43	NF2
LED_STATE/P1.10/NRF52840	2	42	P1.06
P0.03	3	41	NF1
AI4	4	40	P1.04
GND	5	39	SWC → SWCLK/J-LINK
P1.13	6	38	P1.02 → SWDIO/J-LINK
AI0	7	37	SWD
AI5	8	36	P1.00
AI7	9	35	P0.24
AI6	10	34	P0.22
		33	P0.13 → TX-LOG/P0.20/NRF52840
		32	P0.20
		31	D+
		30	P17
		29	D-
		28	P15
		27	VBS
		26	RST → RST/NRF52840

XL1	11	25	DCH
P0.26	12	24	GND
XL2	13	23	VDH
P0.06	14	22	P0.07
AI3	15	21	GND
P0.08	16	20	P12
P1.09	17	19	VCC
AI2	18		

XL1/32.768Khz
SCL/P0.26/NRF52840
XL2/32.768Khz
SDA/P0.06/NRF52840
+3V-BAT
GND
+3V-BAT
GND

J1
DEBUG_DOWNLOAD
+3V-BAT 1
RST/NRF52840 2
TX-LOG/P0.20/NRF52840 3
SWDIO/J-LINK 4
SWCLK/J-LINK 5
GND 6

C5 20pF
XL1/32.768Khz
X1 32.768kHz
XL2/32.768Khz
C11 20pF

+3V-BAT
R4 4.02kΩ
+3V-BAT
R3 4.02kΩ
SCL/P0.26/NRF52840
SDA/P0.06/NRF52840

I²C Sensor Interface — Temperature, Humidity & Pressure

+3V-BAT
C1 10uF

U2 HTU21D
SDA/P0.06/NRF52840
GND
NC
DATA SCK
GND VDD
NC NC
EP GND
SCL/P0.26/NRF52840
+3V-BAT
GND
+3V-BAT
C9 100nF

+3V-BAT
+3V-BAT
C2 100nF

U3 WF100DPZ 0.1BG S6 DT
NC GND
VDD SCL
CSB SDA
GND
SCL/P0.26/NRF52840
SDA/P0.06/NRF52840

HTU21D:
- *Temp + RH sensor*
- *VDD: 1.5 – 3.6 V*
- *I2C Addr: 0x40*
- *I2C: up to 400 kHz*
- *Measuring: 450 µA typ / 500 µA max*
- *Sleep: 0.02 µA typ / 0.14 µA max*
- *Decoupling: 100 nF*

WF100DPZ 0.1BG:
- *Gauge pressure sensor: 10 kPa*
- *VDD: 1.8 – 5.5 V*
- *I2C / SPI (CSB = NC for I2C)*
- *Active current: ≤1 mA design value*
- *Sleep: 0.1 µA typ / 0.3 µA max*
- *Decoupling: 100 nF*
- *I2C address: TBD / verify with manufacturer*

BATTERY - CR2032

+3V-BAT
+3V-BAT
C10 47uF
BAT1 MY-2032-05-R

Supply:
- *VCC / VDDH: 1.7 V to 3.6 V in Normal Voltage Mode*
- *Direct CR2032 supply*
- *VCC pin 19 tied to VDH pin 23*
- *Logic level: 3.3 V nominal*
- *DCH pin 25: NC*

MCU:
- *Nordic nRF52840*
- *ARM Cortex-M4F @ 64 MHz*

Memory:
- *Flash: 1024 KB*
- *RAM: 256 KB*

Wireless:
- *Bluetooth Low Energy 4.2 / 5.0*
- *2.4 GHz ISM*
- *TX power up to +8 dBm*
- *Integrated ceramic antenna*

Clock sources:
- *HFXO: 32 MHz crystal integrated in module*
- *LFCLK: internal LFRC available*
- *Optional external 32.768 kHz crystal:*
  XL1 pin 11 / XL2 pin 13

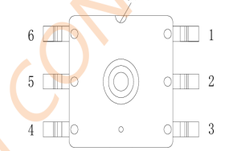
User controls:
- *RESET: RST / P0.18, active LOW*
- *No BOOT button required for SWD programming*

Interfaces:
- *I2C: P0.26 SCL / P0.06 SDA*
- *UART TX LOG: P0.20*
- *LED STATE: P1.10*
- *Debug: SWDIO pin 37 / SWCLK pin 39*

Programming:
- *J-Link / SWD*

Schematic	Schematic1	Create at	2026-07-31
		Update at	2026-07-31
Board	Board1	Page	P1
Drawn		BioSense BLE — Ultra-Low-Power Biodigester Monitoring Node	
Reviewed			
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com

6.CONNECTIONS



Top view

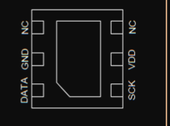
Table2 Definition

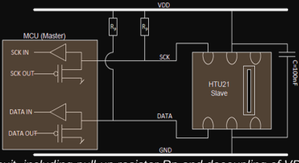
Pin number	Definition	Definition
1	GND	Connected to GND
2	SCL	Clock input for I2C/SPI
3	SDA	(1) Data in/out for I2C. (2) Data input for SPI.
4	CSB	Pulling Low = SPI, NC = IIC
5	VDD	Positive supply voltage
6	NC	NC

Remark：the connections definition is same for SOP and DIP package product

N°	Function	Comment
1	DATA	Data bit-stream
2	GND	Ground
3	NC	Must be left unconnected
4	NC	Must be left unconnected
5	VDD	Supply Voltage
6	SCK	Selector for RH or Temp
PAD		Ground or unconnected







Typical application circuit, including pull-up resistor Rp and decoupling of VDD and GND by a capacitor.